
AB
a/ε
b
④ S → AB
A → a/ε
B → b/ε
FIRST(S) = FIRST(A)
= {a, ε}
FIRST(B) = {b, ε}
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
Ece Department, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026035° Long 80.016136°
Thursday, 15/10/2026 10:14 AM GMT +05:30
Note : Captured by GPS Map Camera
Google

SERVERLESS COMPUTING

Reducing Infrastructure Management in the Cloud

Function-as-a-Service (FaaS) • Architecture • Business Advantages • Practical Use Cases

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SERVERLESS COMPUTING

- A cloud computing model where the cloud provider manages servers and infrastructure.
- Developers focus on writing and deploying application code.
- No direct management of servers, operating systems, or hardware.
- Resources are automatically allocated based on application demand.
- Examples: AWS Lambda, Google Cloud Functions, and Azure Functions.

FUNCTION-AS-A-SERVICE (FaaS)

- FaaS is a serverless execution model.
- Applications are divided into small, independent functions.
- Each function performs a specific task and runs when triggered.
- Functions automatically scale according to workload.
- Example: Image Upload → Function Triggered → Image Resized → Cloud Storage.

SERVERLESS ARCHITECTURE

- User → API Gateway → Cloud Function → Database / Storage
- Client: Sends a request to the application.
- API Gateway: Receives and routes requests.
- FaaS Function: Executes the required business logic.
- Database / Storage: Stores application data.
- Cloud Provider: Handles servers, scaling, and maintenance.
- Key features: Event-driven, stateless, scalable, and server management-free.

BUSINESS ADVANTAGES

- ❑ Reduced Cost: Pay mainly for resources actually used.
- ❑ Automatic Scaling: Handles changes in application traffic.
- ❑ Less Management: No server maintenance or infrastructure administration.
- ❑ Faster Development: Developers focus on application features.
- ❑ High Availability: Cloud providers manage the underlying infrastructure.
- ❑ Faster Deployment: Individual functions can be developed and deployed independently.

PRACTICAL USE CASES & CONCLUSION

- E-Commerce: Order and payment processing.
- Healthcare: Processing medical-record requests.
- Media: Automatic image and video processing.
- Notifications: Email and SMS notifications.
- IoT: Processing sensor data and alerts.
- Conclusion: Serverless computing reduces infrastructure management while providing scalable and efficient application development.
- Platforms: AWS Lambda | Google Cloud Functions | Azure Functions

References: P. Sbarski et al., Serverless Architectures on AWS, Manning Publications; R. Aluvalu & U. Maheswari V. (eds.), Serverless Computing, IGI Global.

THANK YOU